

Secure Exit

SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO2-AT2-Scenario Based

View Only

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QUESTIONS

Q1

Smart Hospital Patient
Registration System
20 marks

Q2

Online Library Book
Management System
20 marks

2/2 answered

Q1

Smart Hospital Patient Registration
System20
marks

Smart Hospital Patient Registration System

A multi-specialty hospital wants to digitize its outpatient registration system.

Every patient arriving at the hospital receives a registration number. The hospital management wants a Java application that maintains patient records efficiently while preventing duplicate registrations.

Each patient record consists of

- ☐ Patient ID
- ☐ Patient Name
- ☐ Mobile Number
- ☐ Department
- ☐ Doctor Name

The system should utilize Java Collection Framework and String manipulation techniques for storing, validating, searching, and displaying patient information.

Functional Requirements

Develop a Java program using ArrayList and HashMap to perform the following operations.

Menu

1. Register Patient
2. Search Patient

3. Display All Patients
4. Count Department-wise Patients
5. Exit

Requirement 1 – Register Patient (4 Marks)

Accept

Patient ID

Patient Name

Mobile Number

Department

Doctor Name

Perform validations.

☐ Patient ID should be unique.

☐ Patient Name should contain only alphabets.

☐ Mobile Number should contain exactly 10 digits.

☐ Department should automatically convert to uppercase.

☐ Remove unnecessary spaces from all string inputs.

Store the patient in

☐ ArrayList

☐ HashMap (PatientID as Key)

Requirement 2 – Search Patient (4 Marks)

Accept Patient ID.

Display complete patient details.

If not found,

Patient Not Found

Requirement 3 – Display All Patients (4 Marks)

Display in the following format.

```
-----  
---  
ID Name Mobile Department Doctor  
-----  
---
```

P101 Rahul 9876543210 CARDIOLOGY Dr.Arun

P102 Priya 9876543211 ORTHO Dr.Meena

Requirement 4 – Department-wise Count (4 Marks)

Count the number of patients in each department.

Example

CARDIOLOGY : 5

ORTHO : 3

NEURO : 4

Use

HashMap<String,Integer>

Requirement 5 – String Operations (4 Marks)

For every patient,

Perform

□ Convert Name to Proper Case

□ Remove leading/trailing spaces

□ Display Name Length

□ Display Department in Uppercase

Example

Input

raHUI kumAr

Output

Rahul Kumar

Length : 11

Department : CARDIOLOGY

Constraints

Maximum Patients = 100

Duplicate Patient IDs are not allowed.

Collections must be used.

Arrays are not allowed.

No Database.

Use only Java Collections.

Sample Input

1

P101

Rahul Kumar

9876543210

Cardiology

Dr.Arun

1

P102

Priya

9876543211

Ortho

Dr.Meena

3

4

2

P101

5

Sample Output

Patient Registered Successfully.

Patient Registered Successfully.

ID Name Department

P101 Rahul Kumar CARDIOLOGY

P102 Priya ORTHO

Department Count

CARDIOLOGY : 1

ORTHO : 1

Patient Details

ID : P101

Name : Rahul Kumar

Mobile : 9876543210

Department : CARDIOLOGY

Doctor : Dr.Arun

Your Code

```
1 import java.util.*;  
2 class Patient{
```

```

3      String id,name,mob,dept,doc;
4
5      Patient(String i,String n,String m,String d,String dc) {
6          id=i;
7          name=proper(n.trim());
8          mob=m.trim();
9          dept=d.trim().toUpperCase();
10         doc=dc.trim();
11     }
12     static String proper(String s) {
13         String[] a=s.toLowerCase().split(' ');
14         String r="";
15         for(String x:a)
16             r=r+Character.toUpperCase(x.charAt(0))+x.substring(1);
17         return r.trim();
18     }
19
20     public static void main(String[] args) {
21         Scanner sc= new Scanner(System.in);
22
23         ArrayList<Patient> list = new ArrayList<>();
24         HashMap<String,Patient> map =new HashMap<>();
25         while(sc.hasNextInt()) {
26             int ch=Integer.parseInt(sc.nextInt());
27             if(ch==1) {
28                 String id=sc.nextLine().trim();
29                 String name=sc.nextLine().trim();
30                 String mob=sc.nextLine().trim();
31                 String dep=sc.nextLine().trim();
32                 String doc=sc.nextLine().trim();
33
34                 if(map.containsKey(id)) {
35                     System.out.println("Duplicate ID");
36                     continue;
37                 }
38                 if(!name.matches("[A-Za-z ]+")) {
39                     System.out.println("Invalid Name");
40                     continue;
41                 }
42                 if(!mob.matches("\\d{10}")) {
43                     System.out.println("Invalid Mobile Number");
44                     continue;
45                 }
46                 Patient p=new Patient(id,name,mob,dep,doc);
47                 list.add(p);
48                 map.put(id,p);
49                 System.out.println("Patient Added Successfully");
50

```

```

51         }
52         else if(ch==2) {
53             String id=sc.nextLine().trim();
54             if(map.containsKey(id)) {
55                 Patient p=map.get(id);
56                 System.out.println("Patient ID: "+id);
57                 System.out.println("Name: "+p.getName());
58                 System.out.println("Age: "+p.getAge());
59                 System.out.println("Mobile: "+p.getMobile());
60                 System.out.println("Department: "+p.getDept());
61                 System.out.println("Doctor: "+p.getDoctor());
62             }
63         }
64         else{
65             System.out.println("Patient not found");
66         }
67     }
68 }
69 else if(ch==3) {
70     System.out.println("-----");
71     System.out.println("ID Name Age Dept Doctor");
72     System.out.println("-----");
73     for(Patient p:list)
74         System.out.println(p.getId()+" "+p.getName()+" "+p.getAge()+" "+p.getDept()+" "+p.getDoctor());
75 }
76 }
77 else if(ch==4) {
78     HashMap<String,Integer> cnt = new HashMap<>();
79     for(Patient p:list)
80         cnt.put(p.getDept(), cnt.getOrDefault(p.getDept(), 0)+1);
81     System.out.println("Department-wise count");
82     for(String k:cnt.keySet())
83         System.out.println(k+" : "+cnt.get(k));
84 }
85 else if(ch==5)
86     break;
87 }
88 }}
89 }

```

Input

```

1
P101
Rahul Kumar
9876543210

```

Output

```
Patient Registered Successfully.  
Patient Registered Successfully.
```

```
-----  
-----  
ID Name Department  
-----
```

Visible Test Cases

Test Case 1



Test Case 2



Test Case 3



0.00 / 20 marks

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